

DIFFERENTIAL EQUATIONS LECTURE NOTES

WEEK (V)

SYSTEM OF FIRST ORDER ODES.

Definition Say there is one independent variable t , and some dependent variable $x_1, x_2, \dots, x_n$. Then, a system of equations in the following form

$$\left. \begin{aligned} x_1' &= f_1(t, x_1, x_2, \dots, x_n) \\ x_2' &= f_2(t, x_1, x_2, \dots, x_n) \\ &\vdots \\ x_n' &= f_n(t, x_1, x_2, \dots, x_n) \end{aligned} \right\} \text{ is called a system of 1st order ODEs.}$$

Ex 7 (1) $\frac{dx_1}{dt} = \underbrace{x_1 + x_2 + 2t}_{f_1(t, x_1, x_2)}$
 $\frac{dx_2}{dt} = \underbrace{x_1 - x_2 t^2}_{f_2(t, x_1, x_2)}$

(2) $x_1' = x_1^2 x_2 + t$
 $x_2' = x_1 - x_2 t.$

Theorem (Existence and Uniqueness)

$$(1) \begin{cases} x_1' = f_1(t, x_1, x_2, \dots, x_n) \\ x_2' = f_2(t, x_1, x_2, \dots, x_n) \\ \vdots \\ x_n' = f_n(t, x_1, x_2, \dots, x_n) \end{cases} \text{ with ICs } \begin{cases} x_1(t_0) = x_1^0 \\ x_2(t_0) = x_2^0 \\ x_3(t_0) = x_3^0 \\ \vdots \\ x_n(t_0) = x_n^0 \end{cases} \quad (2)$$

(1) + (2) defines an I.V.P for system of first order ODEs.

→ If $f_1, f_2, \dots, f_n$ and their partial derivatives $\frac{\partial f_1}{\partial x_1}, \frac{\partial f_2}{\partial x_2}, \dots, \frac{\partial f_1}{\partial x_n},$

$\dots, \frac{\partial f_2}{\partial x_1}, \frac{\partial f_2}{\partial x_n}, \dots, \frac{\partial f_n}{\partial x_1}, \frac{\partial f_n}{\partial x_2}, \dots, \frac{\partial f_n}{\partial x_n}$ are continuous in a region

R containing the initial point $(t_0, x_1^0, \dots, x_n^0)$ then $\exists$ an interval $|t - t_0| < h$ in which $\exists$ a unique solution for the I.V.P (1) + (2)

(c1)

Definition The initial value problem (1)+(2) has a solution if $\exists$ a set of differentiable functions $\{\phi_1(t), \phi_2(t), \dots, \phi_n(t)\}$ satisfying the D.E (1) and the I.C's (2)

Definition 1 A system of 1st order ODEs is "linear" if it can be written in the form

$$x_1' = a_{11}(t)x_1 + a_{12}(t)x_2 + \dots + a_{1n}(t)x_n + g_1(t)$$

$$x_2' = a_{21}(t)x_1 + a_{22}(t)x_2 + \dots + a_{2n}(t)x_n + g_2(t)$$

⋮

$$x_n' = a_{n1}(t)x_1 + a_{n2}(t)x_2 + \dots + a_{nn}(t)x_n + g_n(t)$$

Then, we can write it in matrix-vector form:

$$\begin{bmatrix} x_1' \\ x_2' \\ \vdots \\ x_n' \end{bmatrix} = \underbrace{\begin{bmatrix} a_{11}(t) & a_{12}(t) & \dots & a_{1n}(t) \\ a_{21}(t) & a_{22}(t) & \dots & a_{2n}(t) \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1}(t) & a_{n2}(t) & \dots & a_{nn}(t) \end{bmatrix}}_{A(t)} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} + \begin{bmatrix} g_1(t) \\ g_2(t) \\ \vdots \\ g_n(t) \end{bmatrix}$$

$$\vec{x}' = A(t) \cdot \vec{x} + \vec{g}(t) \quad \text{where} \quad \vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}, \vec{g} = \begin{bmatrix} g_1 \\ g_2 \\ \vdots \\ g_n \end{bmatrix}$$

(2) If $g_i(t) = 0 \quad \forall i = 1, 2, \dots, n$ ($\vec{g}(t) = \vec{0}$), then the system is called homogeneous. Otherwise, that is $\vec{g}(t) \neq \vec{0}$ some of $g_i \neq 0$ for some i , then the system is called nonhomogeneous.

$$\vec{x}' = A(t) \vec{x} + \vec{g}(t) \quad \text{(N)} \quad \text{(non-homogeneous system)}$$

$$\vec{x}' = A(t) \vec{x} \quad \text{(H)} \quad \text{(homogeneous system)}$$

Ex ① $x_1' = 3x_1 - tx_2 + 1$
 $x_2' = tx_1 - x_2$

$\vec{x}' = \underbrace{\begin{bmatrix} 3 & -t \\ t & -1 \end{bmatrix}}_{A(t)} \vec{x} + \underbrace{\begin{bmatrix} 1 \\ 0 \end{bmatrix}}_{\vec{p}(t)}$ non-homogeneous
 1st order linear ODE

$f_1 = 3x_1 - tx_2 + 1$ cont. $\forall t, x_1, x_2 \in \mathbb{R}$

$f_2 = tx_1 - x_2$ cont. $\forall t, x_1, x_2 \in \mathbb{R}$

$\frac{\partial f_1}{\partial x_1} = 3, \frac{\partial f_1}{\partial x_2} = -t, \frac{\partial f_2}{\partial x_1} = t, \frac{\partial f_2}{\partial x_2} = -1$ are all continuous $\forall t, x_1, x_2 \in \mathbb{R}$

Interval of solution $\forall t \in \mathbb{R}$ for any (t_0, x_1^0, x_2^0)

Existence & Uniqueness Thm for System of linear ODEs

If $A(t)$ and $\vec{p}(t)$ are continuous on I , then the I.V.P.

$$\vec{x}'(t) = A(t) \vec{x}(t) + \vec{p}(t)$$

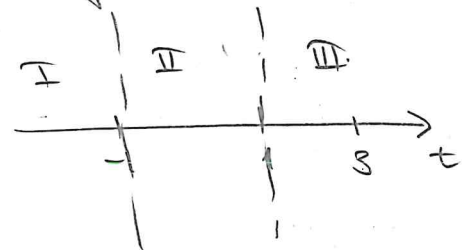
$$\vec{x}(t_0) = \vec{x}_0$$

has a unique solution defined for all $t \in I$.

Ex $\vec{x}'$
 I.V.P. $\begin{cases} x_1' = 3x_1 - tx_2 + \frac{1}{t^2-1} \\ x_2' = tx_1 - x_2 \\ \vec{x}(3) = \begin{bmatrix} 0 \\ -2 \end{bmatrix} \end{cases}$

$\vec{x}'(t) = \underbrace{\begin{bmatrix} 3 & -t \\ t & -1 \end{bmatrix}}_{A(t)} \vec{x} + \underbrace{\begin{bmatrix} \frac{1}{t^2-1} \\ 0 \end{bmatrix}}_{\vec{p}(t)}$

Linear diff.



$A(t)$ is cont. every where $\forall t \in \mathbb{R}$

$\vec{p}(t)$ is continuous $\forall t \in \mathbb{R} - \{ -1, 1 \}$

Interval of solution = $(1, \infty)$ containing initial point

$\therefore$ The system has a unique solution on $I = (1, \infty)$

Remark First consider the homogeneous system

$$\vec{x}' = A(t) \vec{x} \quad (H)$$

Thm If $\vec{x}_1, \vec{x}_2, \dots, \vec{x}_n$ are solutions of (H) $\vec{x}' = A(t) \vec{x}$, it follows that the "linear combination"

$$\vec{x} = c_1 \vec{x}_1 + c_2 \vec{x}_2 + \dots + c_n \vec{x}_n \quad \text{"superposition principle"}$$

is also a solution of (H) for any constant $c_1, c_2, \dots, c_n \in \mathbb{R}$

Definition (1) A set of solutions $\{\vec{x}_1, \vec{x}_2, \dots, \vec{x}_n\}$ of (H) is called linearly independent on I if

$$c_1 \vec{x}_1(t) + c_2 \vec{x}_2(t) + \dots + c_n \vec{x}_n(t) = \vec{0} \quad \forall t \in I$$

implies that $c_1 = c_2 = \dots = c_n = 0$ otherwise, the set is called linearly independent.

(2) A linearly independent set S of solutions of (H) such that every solution of (H) is expressible as a linear combination of elements in S is said to be a basis for the space of solutions of (H). Thus, $S = \{\vec{x}_1, \vec{x}_2, \dots, \vec{x}_n\}$ is a basis for the space of solutions, then this basis is not unique

Thm (1) Any two bases for the same solution space have the same number of elements, which is called the dimension of solution space. (If $A_{n \times n} \Rightarrow$ any basis has n elements)

(2) Any linearly independent set containing n solutions is a basis. Thus, any linear combination gives a solution for the (H).

Thm If $\vec{x}_1, \vec{x}_2, \dots, \vec{x}_n$ are linearly independent solutions of the system

$$\vec{x}' = A(t) \vec{x} \quad (H)$$

for each $t \in I$, then each solution $\vec{x} = \phi(t)$ of (H) can be expressed as a linear combination of $\vec{x}_1, \dots, \vec{x}_n$.

$$\vec{x} = \phi(t) = c_1 \vec{x}_1 + c_2 \vec{x}_2 + \dots + c_n \vec{x}_n$$

in a unique way, and $\phi(t)$ is called a **general solution** of (H).

Definition A linearly ^{independent} set of solutions $\{\vec{x}_1, \vec{x}_2, \dots, \vec{x}_n\}$ is called "**fundamental set of solutions**". The matrix

$X(t) = [\vec{x}_1(t), \vec{x}_2(t), \dots, \vec{x}_n(t)]$ which takes $\vec{x}_p$ as columns is called "**fundamental matrix**".

Properties (1) $\det X(t) \neq 0 \quad \forall t \in I$ since $\vec{x}_p(t)$ are linearly independent

And $\det X(t) = W\{\vec{x}_1, \vec{x}_2, \dots, \vec{x}_n\}$ is called Wronskian of $\vec{x}_1, \vec{x}_2, \dots, \vec{x}_n$.

(2)
$$\begin{aligned} X'(t) &= [\vec{x}'_1(t) \quad \vec{x}'_2(t) \quad \dots \quad \vec{x}'_n(t)] \\ &= \begin{bmatrix} A(t)\vec{x}_1(t) & A(t)\vec{x}_2(t) & \dots & A(t)\vec{x}_n(t) \end{bmatrix} \quad \begin{matrix} \vec{x}' = A\vec{x} \\ \swarrow \text{Since } \vec{x}_p \text{ are solutions of (H).} \end{matrix} \\ &= A(t) [\vec{x}_1(t) \quad \vec{x}_2(t) \quad \dots \quad \vec{x}_n(t)] \end{aligned}$$

$$X'(t) = A(t) X(t)$$

That is, the fundamental matrix X also satisfies the (H) $\vec{x}' = A(t)\vec{x}$

③ $X(t)\vec{c}$, $\vec{c} = \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix}$

$$X(t)\vec{c} = [\vec{x}_1(t) \ \vec{x}_2(t) \ \dots \ \vec{x}_n(t)] \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix} = c_1 \vec{x}_1(t) + c_2 \vec{x}_2(t) + \dots + c_n \vec{x}_n(t)$$

which is a general solution

That is, $X(t)\vec{c}$ is a general solution of (H).

Theorem Let $S = \{ \vec{x}_1(t), \vec{x}_2(t), \dots, \vec{x}_n(t) \}$ be a set of solutions of $\vec{x}'(t) = A(t)\vec{x}(t)$ (H)

defined on I , and $X(t) = [\vec{x}_1(t) \ \vec{x}_2(t) \ \dots \ \vec{x}_n(t)]$. Then

the followings are equivalent.

- ① S is linearly independent
- ② S is a fundamental set of solutions of (H)
- ③ $X(t)$ is a fundamental matrix
- ④ $\det X(t) \neq 0 \quad \forall t \in I \Leftrightarrow \det X(t) = \chi(\vec{x}_1, \vec{x}_2, \dots, \vec{x}_n) \neq 0$ ^(H.1)
- ⑤ $\det X(t) = \chi(\vec{x}_1, \vec{x}_2, \dots, \vec{x}_n) \neq 0$ for some $t \in I$
 (If the wronskian of set of functions is not zero even for one value of t , then the set of functions is linearly independent)
- ⑥ $\vec{x}(t) = X(t) \cdot \vec{c} = c_1 \vec{x}_1(t) + c_2 \vec{x}_2(t) + \dots + c_n \vec{x}_n(t)$ is a general solution of (H).
- ⑦ $\vec{x}(t) = \vec{x}_h(t) + \vec{x}_p(t)$ is general solution of $\vec{x}'(t) = A(t)\vec{x}(t) + \vec{p}(t)$ (N)

where $\vec{x}_h(t) = X(t)\vec{c}$ is the homogeneous (complement)

Solution (i.e. solution of (H)) and $x_p(t)$ is a particular

Solution of (N).

Ex $\vec{x}' = \begin{bmatrix} -1 & 6 \\ 1 & -2 \end{bmatrix} \vec{x}$, $I = (-\infty, \infty)$ since $A(t) = \text{constant matrix}$
 then, it is cont $\forall t \in \mathbb{R}$. Solution exist and
 Unique $\mathbb{R} = (-\infty, \infty)$

Suppose $\vec{x}_1(t) = e^t \begin{bmatrix} 3 \\ 1 \end{bmatrix}$, $\vec{x}_2(t) = e^{-4t} \begin{bmatrix} -2 \\ 1 \end{bmatrix}$ are given.

(a) Show that $\vec{x}_1(t)$ and $\vec{x}_2(t)$ are solutions of the system

$$\begin{aligned} \vec{x}_1' &= (e^t \begin{bmatrix} 3 \\ 1 \end{bmatrix})' = \begin{bmatrix} 3e^t \\ e^t \end{bmatrix} \\ \vec{x}_1' &= A \vec{x}_1 ? \end{aligned} \quad \Rightarrow \quad \begin{aligned} &\vec{x}_1 \text{ satisfies } \vec{x}_1' = A \vec{x}_1 \\ &\text{Therefore, it is a solution.} \end{aligned}$$

$$\begin{bmatrix} 3e^t \\ e^t \end{bmatrix} = \begin{bmatrix} -1 & 6 \\ 1 & -2 \end{bmatrix} \begin{bmatrix} 3e^t \\ e^t \end{bmatrix} = \begin{bmatrix} 3e^t \\ e^t \end{bmatrix}$$

Similarly, $\vec{x}_2'(t) = A(t)\vec{x}_2$ can be shown, $\vec{x}_2$ is a solution (Exerc)

(b) Is $\{\vec{x}_1(t), \vec{x}_2(t)\}$ linearly independent?

Check $W(\vec{x}_1, \vec{x}_2) = \det \begin{bmatrix} 3e^t & -2e^{-4t} \\ e^t & e^{-4t} \end{bmatrix}$ suppose $t=0$

$$W(\vec{x}_1, \vec{x}_2)(t=0) = \begin{vmatrix} 3 & -2 \\ 1 & 1 \end{vmatrix} = 3+2=5 \neq 0$$

by thm $\vec{x}_1(t)$ and $\vec{x}_2(t)$ are linearly independent.

(c) The set $\{\vec{x}_1(t), \vec{x}_2(t)\}$ is a fund. set of solutions of the system since $\vec{x}_1(t)$ and $\vec{x}_2(t)$ are linearly independent by part (b)

(d) $X(t) = \begin{bmatrix} 3e^t & -2e^{-4t} \\ e^t & e^{-4t} \end{bmatrix}$ fundamental matrix

$\underbrace{\qquad\qquad\qquad}_{\vec{x}_1(t)} \quad \underbrace{\qquad\qquad\qquad}_{\vec{x}_2}$

(e) $\vec{x}(t) \in \mathbb{R}^2$ is a general solution that is $\vec{x}(t) = c_1 \underbrace{\begin{bmatrix} 3e^t \\ e^t \end{bmatrix}}_{\vec{x}_1(t)} + c_2 \underbrace{\begin{bmatrix} -2e^{-4t} \\ e^{-4t} \end{bmatrix}}_{\vec{x}_2}$

Ex Find the solution of the I.V.P.

$$\vec{x}' = \begin{bmatrix} -1 & 6 \\ 1 & -2 \end{bmatrix} \vec{x} \quad \vec{x}(0) = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

By previous example, the general solution is

$$\vec{x}(t) = X(t) \cdot \vec{c} = \begin{bmatrix} 3e^t & -2e^{-4t} \\ e^t & e^{-4t} \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix}$$

Put initial value, $t=0 \Rightarrow \vec{x}(0) = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$.

$$\underbrace{\begin{bmatrix} 3 & -2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix}}_{\vec{x}(0)} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\left[\begin{array}{cc|c} 3 & -2 & 0 \\ 1 & 1 & 1 \end{array} \right] \xrightarrow{-3R_2+R_1} \left[\begin{array}{cc|c} 0 & -5 & -3 \\ 1 & 1 & 1 \end{array} \right] \xrightarrow{R_1 \leftrightarrow R_2} \left[\begin{array}{cc|c} 1 & 1 & 1 \\ 0 & -5 & -3 \end{array} \right]$$

$$-5c_2 = -3 \quad c_2 = 3/5.$$

$$c_1 + c_2 = 1$$

$$c_1 = 1 - c_2 = 1 - 3/5 = 2/5.$$

Solution of I.V.P

$$\boxed{\vec{x}(t) = \frac{2}{5} e^t \begin{bmatrix} 3 \\ 1 \end{bmatrix} + \frac{3}{5} e^{-4t} \begin{bmatrix} -2 \\ 1 \end{bmatrix}}$$

HOW TO FIND SOLUTIONS OF 1ST ORDER LINEAR SYSTEM OF ODES.

→ We'll consider only the case where A is a constant matrix.

$$\begin{aligned}\vec{x}'(t) &= A(t)\vec{x} + \vec{p}(t) & (N) \\ \vec{x}'(t) &= A(t)\vec{x} & (H)\end{aligned}$$

when A is constant, then we have

$$\begin{aligned}\vec{x}'(t) &= A\vec{x}(t) + \vec{p}(t) & (\tilde{N}) \\ \vec{x}'(t) &= A\vec{x}(t) & (\tilde{H})\end{aligned}$$

We will consider the solution of linear homogeneous 1st order system with A is constant.

• Now, consider only one O.D.E, i.e.,

$$\begin{aligned}x' &= ax \Rightarrow a \text{ constant "separable"} \\ \frac{dx}{dt} &= a \Rightarrow \ln|x| = at + c_1 \Rightarrow x(t) = \pm e^{c_1} e^{at} \\ & \qquad \qquad \qquad x(t) = c e^{at}\end{aligned}$$

[Since we need function whose derivative is proportional to] themselves, we use exponential functions
→ Thus, we may look for a solution of $(\tilde{H}) \vec{x}'(t) = A\vec{x}(t)$

in the form $\vec{x} = e^{\lambda t} \vec{v}$, $\lambda \in \mathbb{C}$ and $\vec{v} \neq 0$ constant vector

→ Clearly, $\vec{x}' = \lambda e^{\lambda t} \vec{v}$, substitute in $(\tilde{H})$ then, we get

$$\vec{x}' = A\vec{x}$$

$$\lambda e^{\lambda t} \vec{v} = A e^{\lambda t} \vec{v}$$

$$A\vec{v} = \lambda \vec{v}$$

"eigenvalue problem"

λ : eigenvalue of A

$\vec{v} \neq 0$ are the corresponding

$$(A - \lambda I)\vec{v} = 0 \text{ "homogeneous algebraic equations of } A."$$

system, which has non-zero solutions when $\det(A - \lambda I) = 0$ "characteristic polynomial"

$\det(A - \lambda I) = n^{\text{th}}$ degree polynomial in λ .

For a polynomial we have three cases for the roots.

- ① λ 's are real distinct
- ② λ 's are complex conjugates
- ③ λ 's are repeated

Remark If λ is an eigenvalue of A with a corresponding eigenvector $\vec{v} \neq 0$, then $e^{\lambda t} \vec{v}$ is a solution of $\vec{x}' = A \vec{x}$ ($\vec{v}$)

Case I: DISTINCT AND REAL EIGENVALUES

$\lambda_1, \lambda_2, \dots, \lambda_n$ eigenvalues of A which are real and distinct, then each corresponds to an eigenvector $\vec{v}_i$ s.t. they are linearly independent $i=1, 2, \dots, n$. Then,

$\{e^{\lambda_1 t} \vec{v}_1, e^{\lambda_2 t} \vec{v}_2, \dots, e^{\lambda_n t} \vec{v}_n\}$ is the set of fundamental

Solutions for $\vec{x}' = A \vec{x}$

Ex $\begin{cases} x_1' = -x_1 + 2x_2 \\ x_2' = -3x_2 \end{cases}$ Solve $\begin{bmatrix} x_1' \\ x_2' \end{bmatrix} = \begin{bmatrix} -1 & 2 \\ 0 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$

For the solution find eigenvalues and eigenvectors of A .

Eigenvalues $\det(A - \lambda I) = 0$

$$\begin{vmatrix} -1-\lambda & 2 \\ 0 & -3-\lambda \end{vmatrix} = (-1-\lambda)(-3-\lambda) = 0 \Rightarrow \lambda_1 = -1, \lambda_2 = -3.$$

Then, find eigenvectors

$$\underline{\lambda = -1} \quad (A - \lambda I)\vec{v} = \vec{0} \quad (A + I)\vec{v} = \vec{0} \quad \vec{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 2 \\ 0 & -2 \end{bmatrix} \xrightarrow{\text{row operation}} \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \quad \begin{array}{l} v_2: \text{leading} \\ v_1: \text{free} \\ v_2 = 0 \end{array}$$

$\text{alp}(\lambda = -1) = 1 = \text{geometric mult. on } (\lambda = -1) = 1 \text{ free} = \neq \text{linearly independent eigenvectors}$

$$\vec{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} v_1 \\ 0 \end{bmatrix} = v_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} \Rightarrow V_{\lambda = -1} = \text{span} \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right\} = \{ \text{vectors of form } c \begin{bmatrix} 1 \\ 0 \end{bmatrix} \}$$

nonzero constant multiple of $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$

Let $v_1 = 1$ v.l.o.b. multiple of $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$

$$\text{So, } \underline{\vec{x}_1(t) = e^{\lambda_1 t} \vec{v} = e^{-t} \begin{bmatrix} 1 \\ 0 \end{bmatrix}}$$

$$\underline{\lambda_2 = -3} \quad (A + 3I)\vec{u} = \vec{0} \quad \vec{u} = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 2 \\ 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \quad \begin{array}{l} u_1: \text{leading} \\ u_2: \text{free} \end{array}$$

$u_1 + u_2 = 0 \Rightarrow u_1 = -u_2$

$\vec{u} = \begin{bmatrix} -u_2 \\ u_2 \end{bmatrix} = u_2 \begin{bmatrix} -1 \\ 1 \end{bmatrix}$

$m(\lambda = -3) = 1 = \text{alp}(\lambda = -3)$

$$V_{\lambda = -3} = \text{span} \left\{ \begin{bmatrix} -1 \\ 1 \end{bmatrix} \right\}$$

solution, $\vec{x}_2(t) = e^{\lambda_2 t} \vec{u} = e^{-3t} \begin{bmatrix} -1 \\ 1 \end{bmatrix}$ we know that the general solution

$$\vec{x}_g = c_1 \vec{x}_1(t) + c_2 \vec{x}_2(t) = c_1 e^{-t} \begin{bmatrix} 1 \\ 0 \end{bmatrix} + c_2 e^{-3t} \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

$$\underline{\text{O.E.}} \quad X(t) = [\vec{x}_1(t) \quad \vec{x}_2(t)] = \begin{bmatrix} e^{-t} & -e^{-3t} \\ 0 & e^{-3t} \end{bmatrix} \quad \text{"fundamental matrix"}$$

$$\vec{x}_g = X(t) \cdot \vec{c} = \begin{bmatrix} e^{-t} & -e^{-3t} \\ 0 & e^{-3t} \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = c_1 e^{-t} \begin{bmatrix} 1 \\ 0 \end{bmatrix} + c_2 e^{-3t} \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

Moreover, A is diagonalizable, since $\exists$ an invertible matrix P such that

$$P = [\vec{v} \quad \vec{u}] = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} \text{ with } J = P^{-1}AP = \begin{bmatrix} -1 & 0 \\ 0 & -3 \end{bmatrix}$$

$$\text{or } P = [\vec{u} \quad \vec{v}] = \begin{bmatrix} -1 & 1 \\ 1 & 0 \end{bmatrix} \text{ with } J = P^{-1}AP = \begin{bmatrix} -3 & 0 \\ 0 & -1 \end{bmatrix}$$

$$\underline{A^{10}} ? \quad J = P^{-1}AP \Rightarrow A = PJ P^{-1}$$

$$A^n = \underbrace{(PJ P^{-1}) (PJ P^{-1}) \dots (PJ P^{-1})}_{n \text{ times}}$$

$$A^n = P \cdot J^n \cdot P^{-1}$$

$$J = \begin{bmatrix} \lambda_1 & & 0 \\ & \lambda_2 & \\ 0 & & \lambda_n \end{bmatrix} \Rightarrow J^n = \begin{bmatrix} \lambda_1^n & & 0 \\ & \lambda_2^n & \\ 0 & & \lambda_n^n \end{bmatrix}$$

$$\therefore A^{10} = PJ^{10}P^{-1} \text{ since } A \text{ is diagonalizable.}$$

$$A^{10} = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} (-1)^{10} & 0 \\ 0 & (-3)^{10} \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

$$\left(\text{Remember: } A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \quad A^{-1} = \frac{1}{\det A} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} \right)$$

$$P = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} \quad P^{-1} = \frac{1}{1} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

How to sketch the solution on x_1, x_2 system (Phase Portrait)

Draw Phase Portrait

$$\vec{x}(t) = \underbrace{c_1 \cdot e^{-t} \begin{bmatrix} 1 \\ 0 \end{bmatrix}}_{\text{dominant as } t \rightarrow \infty} + \underbrace{c_2 e^{-3t} \begin{bmatrix} -1 \\ 1 \end{bmatrix}}_{\text{dominant as } t \rightarrow -\infty}$$

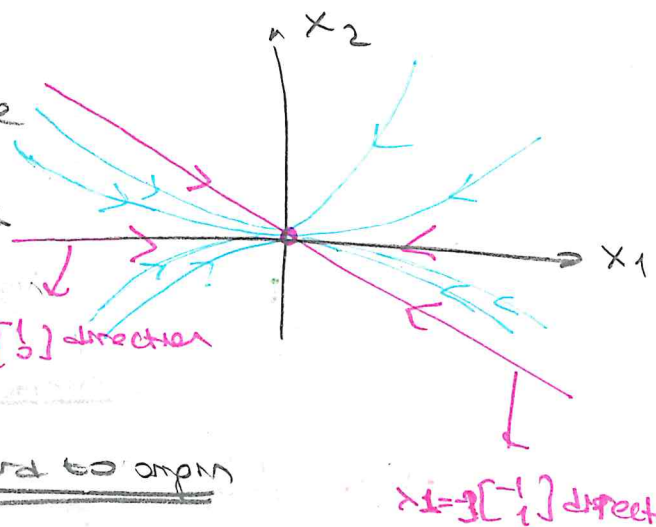
$$t \rightarrow \infty \quad \vec{x}(t) \rightarrow \vec{0}$$

$$t \rightarrow -\infty \quad \vec{x}(t) \rightarrow \vec{0}$$

- ① Draw two straight lines in the direction of eigenvectors, draw arrows on this eigenvectors

NOTE The sign of eigenvalue determine the direction of arrows

- If $\lambda < 0 \Rightarrow$ arrow will be toward to origin
- If $\lambda > 0 \Rightarrow$ arrow will be in the negative direction according to the origin



- ② Draw general trajectories: Since both eigenvalues negative

$\lambda_1 = -3$, $\lambda_2 = -1$, Near the origin the motion look like x_2 ($|\lambda_2 = -1| < |\lambda_1 = -3|$) because $\lambda_1 = -3$ dies first

Therefore, general trajectories will be tangent to $v_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

- ③ Arrow directions on curved trajectories: since both eigenvalues are negative they will approach origin

Remark Suppose the system with 2 solutions

$$\vec{x}(t) = c_1 e^t \begin{bmatrix} 1 \\ 5 \end{bmatrix} + c_2 e^{3t} \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

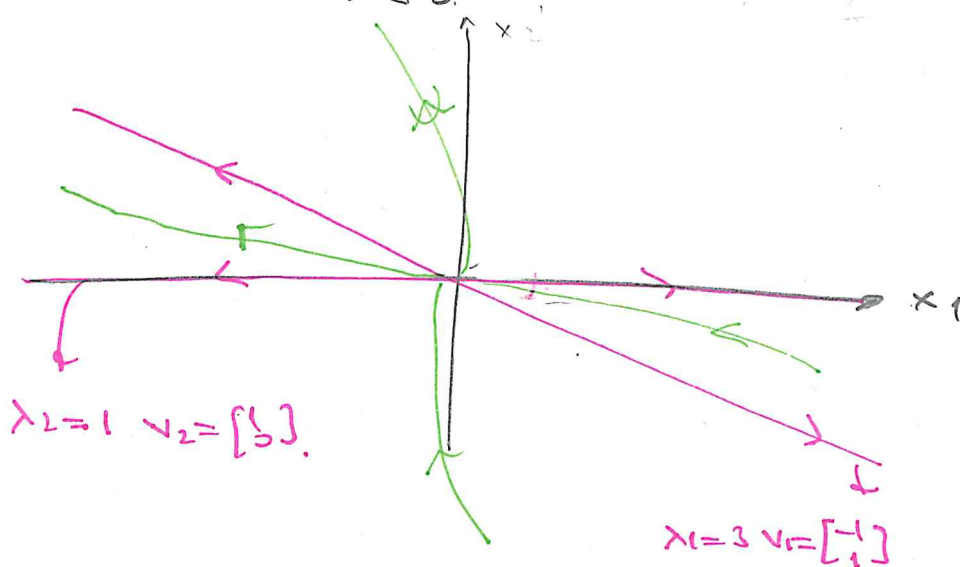
$$\lambda_2 = 1 \quad v_2 = \begin{bmatrix} 1 \\ 5 \end{bmatrix}$$

$$\lambda_1 = 3 \quad v_1 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

$t \rightarrow -\infty$ dominant term (approach 0)
 $t \rightarrow \infty$ dominant term (goes infinity)

$t \rightarrow \infty \quad x(t) \rightarrow \infty$ with direction v_1

$t \rightarrow -\infty \quad x(t) \rightarrow 0$ with direction v_2



Since for $t \rightarrow \infty$ dominant eigenvalue is $\lambda = 3$, this means that trajectories will be parallel to v_1 . Repeat to $|\lambda_2| < |\lambda_1|$ near origin, parallel to v_2 as $t \rightarrow -\infty$.

Remark Origin is the equilibrium solution of $\vec{x}' = Ax$

Origin is called a **node** and it is called **stable node** (nodal sink) when all solutions are going to origin as $t \rightarrow \infty$, as in the first case.

On the other hand, if the solutions go away from origin as $t \rightarrow \infty$, then the origin is called **unstable node** (nodal source)

Ex 7 $\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 5 & -1 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$

Solve and draw phase portrait

$$\det(A - \lambda I) = 0 \quad \begin{vmatrix} 5-\lambda & -1 \\ 3 & 1-\lambda \end{vmatrix} = (5-\lambda)(1-\lambda) + 3$$

$$= \lambda^2 - 6\lambda + 8 = (\lambda-4)(\lambda-2) = 0$$

$\lambda_1 = 2, \lambda_2 = 4$ both positive

$\lambda_1 = 2$ $(A - 2I)\vec{v} = 0 \quad \vec{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$

$$\begin{bmatrix} 3 & -1 \\ 3 & -1 \end{bmatrix} \rightarrow \begin{bmatrix} 3 & -1 \\ 0 & 0 \end{bmatrix} \quad \begin{array}{l} v_1 \text{ is leading} \\ v_2 \text{ is free} \end{array}$$

$$\begin{array}{l} 3v_1 - v_2 = 0 \\ 3v_1 = v_2 \end{array}$$

x1.c.o.n choose $v_2 = 3, v_1 = 1 \quad \vec{v} = \begin{bmatrix} 1/3 \\ 1 \end{bmatrix} v_2$

$$\vec{x}_1(t) = e^{2t} \begin{bmatrix} 1 \\ 3 \end{bmatrix} \quad V_{\lambda=2} = \text{span} \left\{ \begin{bmatrix} 1 \\ 3 \end{bmatrix} \right\}$$

$\lambda_2 = 4$ $(A - 4I)\vec{u} = 0 \quad \vec{u} = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$

$$\begin{bmatrix} 1 & -1 \\ 3 & -3 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix} \quad \begin{array}{l} u_1 \text{ is leading} \\ u_2 \text{ is free} \end{array}$$

$$\begin{array}{l} u_1 - u_2 = 0 \Rightarrow u_1 = u_2 \\ \vec{u} = u_2 \begin{bmatrix} 1 \\ 1 \end{bmatrix} \end{array}$$

$V_{\lambda=4} = \text{span} \left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right\}$

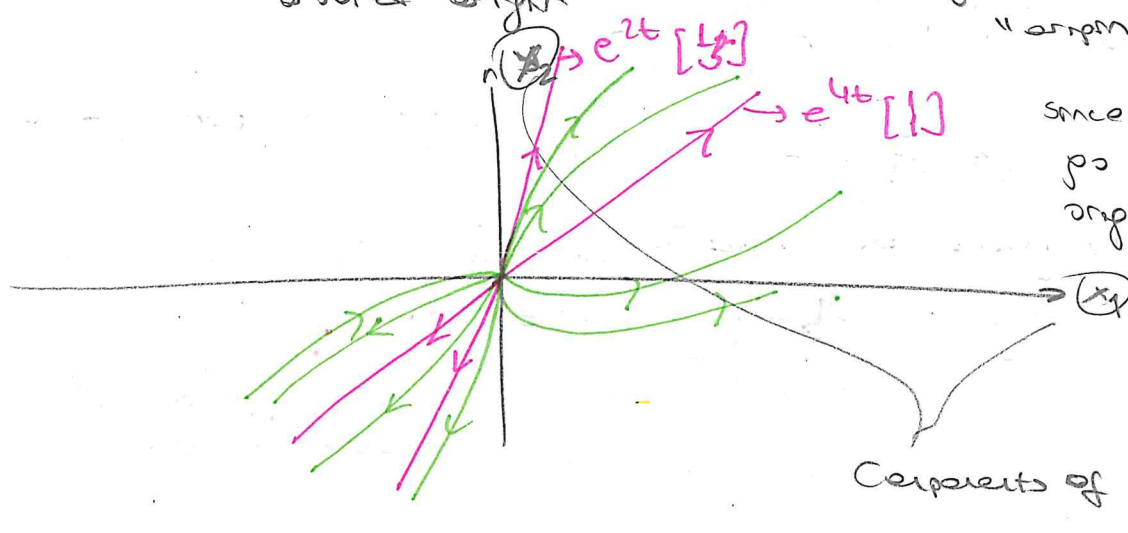
$$\vec{x}_{\text{general}} = C_1 e^{2t} \begin{bmatrix} 1 \\ 3 \end{bmatrix} + C_2 e^{4t} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\begin{array}{l} t \rightarrow \infty \quad x(t) \rightarrow \infty \\ t \rightarrow -\infty \quad x(t) \rightarrow \text{origin} \end{array}$$

denient $t \rightarrow -\infty$ around origin
denient as $t \rightarrow \infty$ far from origin

"origin is unstable node"

since all solutions go away from origin as $t \rightarrow \infty$



Components of $\vec{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$

Ex 7 $\vec{x}' = \begin{bmatrix} 2 & 0 \\ -1 & -4 \end{bmatrix} \vec{x}$ Find sol and draw phase portrait

Eigenvalues $\det(A - \lambda I) = 0$

$$\begin{vmatrix} 2-\lambda & 0 \\ -1 & -4-\lambda \end{vmatrix} = (2-\lambda)(-4-\lambda) = 0 \Rightarrow \lambda_1 = 2 > 0, \lambda_2 = -4 < 0$$

real and distinct.

Eigenvectors $(A - \lambda I)\vec{v} = \vec{0}$

$\lambda = 2$ $(A - 2I)\vec{v} = \vec{0}$ $\vec{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$ $\vec{v} = \begin{bmatrix} -b \\ 1 \end{bmatrix}$ $V_{\lambda=2} = \text{span} \left\{ \begin{bmatrix} -b \\ 1 \end{bmatrix} \right\}$

$$\begin{bmatrix} 0 & 0 \\ -1 & -6 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 6 \\ 0 & 0 \end{bmatrix} \begin{matrix} v_1 = \text{leading} \\ v_2 = \text{free} \end{matrix}$$

$v_1 + 6v_2 = 0 \Rightarrow v_1 = -6v_2$

$\lambda = -4$ $(A + 4I)\vec{v} = \vec{0}$ $\vec{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$

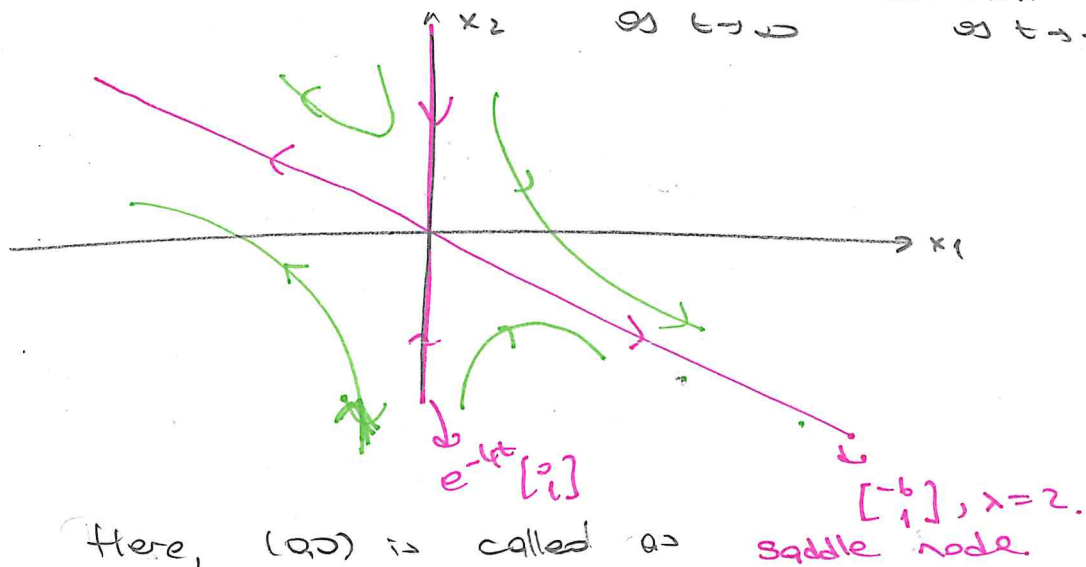
$$\begin{bmatrix} 6 & 0 \\ -1 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{matrix} v_1 = \text{leading} \\ v_2 = \text{free} \end{matrix}$$

$v_1 = 0 \Rightarrow \vec{v} = \begin{bmatrix} 0 \\ v_2 \end{bmatrix} = v_2 \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

$V_{\lambda=-4} = \text{span} \left\{ \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}$

General solution $\vec{x}_{\text{general}} = C_1 \vec{x}_1(t) + C_2 \vec{x}_2(t) = C_1 e^{2t} \begin{bmatrix} -b \\ 1 \end{bmatrix} + C_2 e^{-4t} \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

dominant as $t \rightarrow \infty$ dominant as $t \rightarrow -\infty$



SUMMARY

$$\mathbf{x}' = \mathbf{A}\mathbf{x} \quad \mathbf{A} \in \mathbb{R}^{2 \times 2} \quad \mathbf{A}\mathbf{v} = \lambda \mathbf{v} \quad \lambda_1 < \lambda_2$$

REAL EIGENVALUES

Case	Eigenvalues	Stability	Trajectory near origin	Parallel away from origin
Unstable node (source)	$\lambda_1, \lambda_2 > 0$	Move away from origin	Smaller eigenvalue (v_1)	Larger eigenvalue (v_2)
Stable node (sink)	$\lambda_1, \lambda_2 < 0$	Move toward origin	Less negative (v_2) $(\lambda_2 < \lambda_1)$	Smaller one (more negative v_1)
Saddle point	$\lambda_1 < 0 < \lambda_2$	One stable one unstable	v_1 (stable)	v_2 (unstable)

Result Trajectory to smaller in absolute value, parallel to bigger in absolute value.

CASE II COMPLEX EIGENVALUES

Let $\lambda = \alpha + i\beta$ ($\alpha = \operatorname{Re}(\lambda)$, $\beta = \operatorname{Im}(\lambda)$, $i^2 = -1$) be an eigenvalue with

eigenvector $\vec{u} = \vec{u} + i\vec{v}$. Then,

$$\overline{A\vec{u}} = \overline{\lambda\vec{u}} \Rightarrow \overline{A}\overline{\vec{u}} = \overline{\lambda}\overline{\vec{u}}$$

$$\begin{cases} z = \alpha + i\beta \\ \bar{z} = \alpha - i\beta \end{cases}$$

Since A is real matrix, then $\overline{A} = A$, so

$$A\overline{\vec{u}} = \overline{\lambda}\overline{\vec{u}}$$

Thus $\overline{\lambda} = \alpha - i\beta$ is an eigenvalue of A with the eigenvector

$$\overline{\vec{u}} = \vec{u} - i\vec{v}$$

Fact If $\lambda = \alpha + i\beta$ is complex eigenvalue with corresponding eigenvector $\vec{u} = \vec{u} + i\vec{v}$ then its complex conjugate $\overline{\lambda} = \alpha - i\beta$ is also an eigenvalue with eigenvector $\overline{\vec{u}} = \vec{u} - i\vec{v}$

Thus, $e^{\lambda t}\vec{u}$ and $e^{\overline{\lambda} t}\overline{\vec{u}}$ are two linearly independent solutions

for $\mathbf{x}' = \mathbf{A}\mathbf{x}$ which are complex solutions. So,

$$\vec{x}_p = c_1 \underbrace{e^{\lambda t}\vec{u}}_{\vec{x}_1} + c_2 \underbrace{e^{\overline{\lambda} t}\overline{\vec{u}}}_{\vec{x}_2}$$

Remark In this course, we are only looking for real solutions

For real solutions let's take some suitable linear combinations of these solutions $\vec{x}_1$ and $\vec{x}_2$ ($\vec{x}_2 = \overline{\vec{x}_1}$)

$$c_1 = \frac{1}{2} = c_2 \quad \phi_1(t) = \frac{\vec{x}_1 + \vec{x}_2}{2} = \operatorname{Re}(\vec{x}_1) = \operatorname{Re}(e^{\lambda t} \vec{w}) \in \mathbb{R}$$

which is also a real:

$$c_1 = \frac{1}{2i} \quad c_2 = -\frac{1}{2i} \quad \phi_2(t) = \frac{\vec{x}_1 - \vec{x}_2}{2i} = \operatorname{Im}(\vec{x}_1) = \operatorname{Im}(e^{\lambda t} \vec{w}) \in \mathbb{R}$$

which is also a solution

Now look at $e^{\lambda t} \vec{w} = e^{(\alpha + i\beta)t} (\vec{u} + i\vec{v}) = e^{\alpha t} e^{i\beta t} (\vec{u} + i\vec{v})$

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots \quad x = i\theta$$

$$e^{i\theta} = 1 + i\theta - \frac{\theta^2}{2!} - i\frac{\theta^3}{3!} + \frac{\theta^4}{4!} + i\frac{\theta^5}{5!} - \frac{\theta^6}{6!} - i\frac{\theta^7}{7!} - \dots$$

$$= \left(1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \frac{\theta^6}{6!} + \dots\right) + i\left(\theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \frac{\theta^7}{7!} + \dots\right)$$

$\cos \theta$ $\sin \theta$

$$e^{i\theta} = \cos \theta + i \sin \theta \quad \text{"Euler's Formula"}$$

$$e^{\lambda t} \vec{w} = e^{\alpha t} e^{i\beta t} (\vec{u} + i\vec{v}) = e^{\alpha t} (\cos \beta t + i \sin \beta t) (\vec{u} + i\vec{v})$$

$\cos \beta t + i \sin \beta t$

$$= e^{\alpha t} (\cos \beta t \vec{u} - \sin \beta t \vec{v}) + i e^{\alpha t} (\cos \beta t \vec{v} + \sin \beta t \vec{u})$$

$\phi_1(t) = \operatorname{Re}(e^{\lambda t} \vec{w})$ $\phi_2(t) = \operatorname{Im}(e^{\lambda t} \vec{w})$

Thus, real general solution

$$\vec{x}_p = c_1 \phi_1(t) + c_2 \phi_2(t)$$

$$= c_1 \underbrace{e^{\alpha t} (\cos \beta t \vec{u} - \sin \beta t \vec{v})}_{\phi_1 = \vec{x}_1} + c_2 \underbrace{e^{\alpha t} (\cos \beta t \vec{v} + \sin \beta t \vec{u})}_{\phi_2 = \vec{x}_2}$$

(12)

$$\vec{x}_1 = e^{\alpha t} \begin{bmatrix} \cos \beta t u_1 - \sin \beta t v_1 \\ \cos \beta t u_2 - \sin \beta t v_2 \end{bmatrix} \quad \vec{x}_2 = e^{\alpha t} \begin{bmatrix} \sin \beta t u_1 + \cos \beta t v_1 \\ \sin \beta t u_2 + \cos \beta t v_2 \end{bmatrix}$$

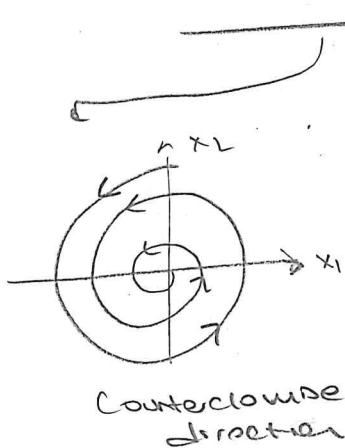
$$\vec{x}_1^2 + \vec{x}_2^2 = e^{2\alpha t} \begin{bmatrix} \cos^2 \beta t u_1^2 + \sin^2 \beta t v_1^2 - 2 \cos \beta t \sin \beta t u_1 v_1 \\ \cos^2 \beta t u_2^2 + \sin^2 \beta t v_2^2 - 2 \cos \beta t \sin \beta t u_2 v_2 \end{bmatrix} + e^{2\alpha t} \begin{bmatrix} \sin^2 \beta t u_1^2 + \cos^2 \beta t v_1^2 + 2 \sin \beta t \cos \beta t u_1 v_1 \\ \sin^2 \beta t u_2^2 + \cos^2 \beta t v_2^2 + 2 \sin \beta t \cos \beta t u_2 v_2 \end{bmatrix}$$

$$\vec{x}_1^2 + \vec{x}_2^2 = e^{2\alpha t} \begin{bmatrix} u_1^2 + v_1^2 \\ u_2^2 + v_2^2 \end{bmatrix} \quad \left\{ \begin{array}{l} \text{NOTE If we use period solution} \\ x_p = \begin{bmatrix} x_1^g \\ x_2^g \end{bmatrix} \quad \begin{array}{l} x_1^g = e^{\alpha t} (K_1 u_1 + K_2 v_1) \\ x_2^g = e^{\alpha t} (K_1 u_2 + K_2 v_2) \end{array} \\ L_1 = C_1 \cos \beta t + C_2 \sin \beta t \\ L_2 = -C_1 \sin \beta t + C_2 \cos \beta t \end{array} \right. \quad \left. \begin{array}{l} \text{we can show } A(x_1^g)^2 + 2B x_1^g x_2^g + C(x_2^g)^2 = D e^{2\alpha t} (C_1^2 + C_2^2) \\ \text{is an ellipse in phase portrait} \end{array} \right.$$

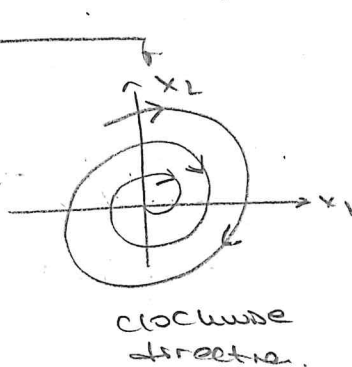
$x_1^2 + y_1^2 = e^{2\alpha t} (u_1^2 + v_1^2)$
 $x_2^2 + y_2^2 = e^{2\alpha t} (u_2^2 + v_2^2)$ } ellipses centered at (0,0) if $\alpha \neq 0$
 with increasing ($\alpha > 0$) or decreasing ($\alpha < 0$) radius.
 ellipses centered (0,0) if $\alpha = 0$
 with fixed radius.

Phase Portrait

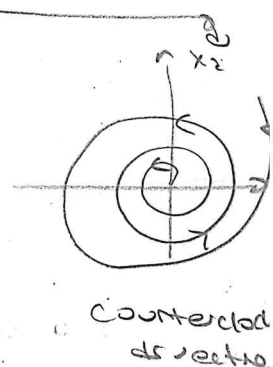
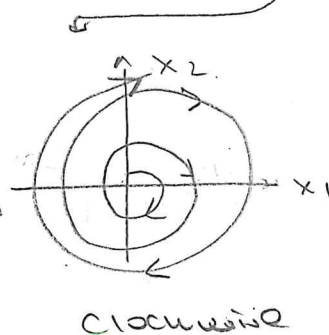
$$\alpha = \operatorname{Re}(\lambda) < 0$$



x_1, x_2 are components of $\vec{x}' = A\vec{x}$ $\vec{x} = [x_1, x_2]^T$



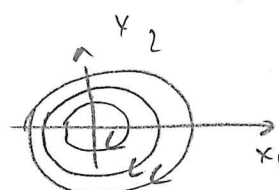
$$\alpha = \operatorname{Re}(\lambda) > 0$$



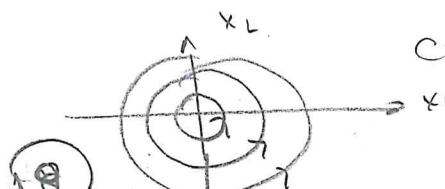
origin is "stable spiral"

origin is "unstable spiral"

$$\alpha = \operatorname{Re}(\lambda) = 0$$



origin is "center" called "neutrally stable"



Phase Portrait

We know it is a stable node.

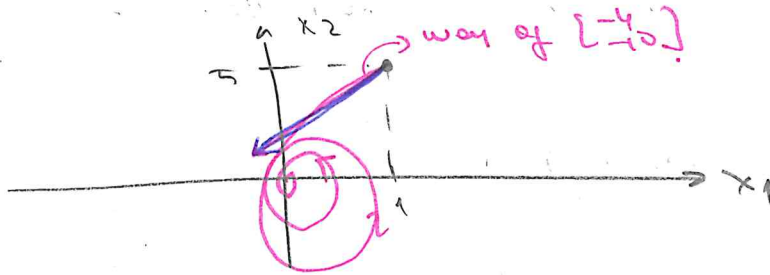
To decide counter clockwise / clockwise, choose $\vec{x} = \begin{bmatrix} 1 \\ 5 \end{bmatrix}$

$$A\vec{x} = \begin{bmatrix} 1 & -1 \\ 5 & -3 \end{bmatrix} \begin{bmatrix} 1 \\ 5 \end{bmatrix} = \begin{bmatrix} -4 \\ -10 \end{bmatrix} \text{ speed vector}$$

$$\begin{aligned} \vec{x}' &= A\vec{x} & \vec{x}(t) - \vec{x}(0) &\approx A\vec{x}_0 \\ \vec{x}(t) &= \vec{x}_0 + A\vec{x}_0 & \vec{x}(t) &= \begin{bmatrix} 1 \\ 5 \end{bmatrix} + \begin{bmatrix} -4 \\ -10 \end{bmatrix} \\ & & & \vec{x}_0 + A\vec{x}_0 \end{aligned}$$

$$\text{since } \text{Re}(\lambda) = -1 < 0$$

stable spiral with positive direction
counter clockwise



$$\begin{aligned} \vec{x}' &= \\ \vec{x}(t) &= \vec{x}_0 \end{aligned}$$

$$\vec{x}(t) =$$

Ex $\vec{x}' = \begin{bmatrix} 1 & 2 \\ -5 & -1 \end{bmatrix} \vec{x}$

$$\det(A - \lambda I) = \begin{vmatrix} 1-\lambda & 2 \\ -5 & -1-\lambda \end{vmatrix} = \lambda^2 + 9 = 0$$

$$\lambda = \pm 3i \quad \alpha + i\beta$$

$$\text{Re}(\lambda) = 0$$

$$\lambda = 3i \quad (A - 3iI)\vec{v} = \vec{0} \quad \vec{v} = \begin{bmatrix} 9 \\ 6 \end{bmatrix}$$

$$\begin{bmatrix} 1-3i & 2 \\ -5 & -1-3i \end{bmatrix} \xrightarrow{R_2 \rightarrow \frac{1+3i}{2}R_1 + R_2} \begin{bmatrix} 1-3i & 2 \\ 0 & 0 \end{bmatrix} \quad \begin{array}{l} a: \text{leading} \\ b: \text{free} \end{array}$$

$$\vec{x}(t) = e^{\lambda t} \vec{v} = e^{3it} \left(\begin{bmatrix} -2 \\ 1 \end{bmatrix} + i \begin{bmatrix} 0 \\ -3 \end{bmatrix} \right)$$

$$= (\cos 3t + i \sin 3t) \left(\begin{bmatrix} -2 \\ 1 \end{bmatrix} + i \begin{bmatrix} 0 \\ -3 \end{bmatrix} \right)$$

$$= (\cos 3t \begin{bmatrix} -2 \\ 1 \end{bmatrix} + \sin 3t \begin{bmatrix} 0 \\ -3 \end{bmatrix}) + i (\cos 3t \begin{bmatrix} 0 \\ -3 \end{bmatrix} + \sin 3t \begin{bmatrix} -2 \\ 1 \end{bmatrix})$$

$$= \begin{bmatrix} -2 \cos 3t \\ 3 \sin 3t + \cos 3t \end{bmatrix} + i \begin{bmatrix} -2 \sin 3t \\ -3 \cos 3t + \sin 3t \end{bmatrix}$$

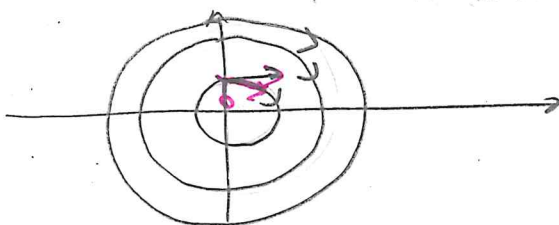
$$\underbrace{\begin{bmatrix} -2 \cos 3t \\ 3 \sin 3t + \cos 3t \end{bmatrix}}_{\vec{x}_1(t)}$$

$$\underbrace{\begin{bmatrix} -2 \sin 3t \\ -3 \cos 3t + \sin 3t \end{bmatrix}}_{\vec{x}_2(t)}$$

Real solution

$$\vec{x}_{\text{real}}(t) = c_1 \begin{bmatrix} -2 \cos 3t \\ 3 \sin 3t + \cos 3t \end{bmatrix} + c_2 \begin{bmatrix} -2 \sin 3t \\ -3 \cos 3t + \sin 3t \end{bmatrix}$$

$$\vec{x}_1^2 + \vec{x}_2^2 = \begin{bmatrix} 4 \\ 10 \end{bmatrix}$$



Decide about way choose $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$

$$A \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ -5 & -1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$$

$$A \begin{bmatrix} 1 \\ 5 \end{bmatrix} = \begin{bmatrix} 1 \\ -5 \end{bmatrix}$$

Origin is center, which is neutrally stable

Ex (H.W) Solve draw phase portrait

a) $\vec{X}' = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \vec{x}$

b) $\vec{X}' = \begin{bmatrix} 2 & -1 \\ 1 & 2 \end{bmatrix} \vec{x}$

Ex $\vec{X}' = \begin{bmatrix} -5 & 5 & 4 \\ -8 & 7 & 6 \\ 1 & 0 & 0 \end{bmatrix} \vec{x}$

$A - \lambda I = \begin{bmatrix} -5-\lambda & 5 & 4 \\ -8 & 7-\lambda & 6 \\ 1 & 0 & -\lambda \end{bmatrix}$

$\det(A - \lambda I) = (-5-\lambda)(-\lambda)(7-\lambda) - 5(8\lambda - 6) + 4(\lambda - 7)$
 $= -(\lambda - 2)(\lambda^2 + 1)$

$\lambda_1 = 2$ $\lambda_{2,3} = \pm i$

$\lambda_1 = 2 \quad (A - 2I) \vec{v} = \vec{0}$

$\begin{bmatrix} -7 & 5 & 4 \\ -8 & 5 & 6 \\ 1 & 0 & -2 \end{bmatrix} \xrightarrow{\text{row}} \begin{bmatrix} 1 & 0 & -2 \\ 0 & 1 & -2 \\ 0 & 0 & 0 \end{bmatrix}$ $v_1, v_2: \text{leading}$
 $v_3: \text{free}$

$v_1 - 2v_3 = 0$ $v_3 = \text{free}$
 $v_2 - 2v_3 = 0$

$\vec{v} = \vec{v}_3 \begin{bmatrix} 2 \\ 2 \\ 1 \end{bmatrix}$ $\vec{v}_{\lambda=2} = \text{span} \left\{ \begin{bmatrix} 2 \\ 2 \\ 1 \end{bmatrix} \right\}$

$\vec{x}_1(t) = e^{\lambda t} \vec{v} = e^{2t} \begin{bmatrix} 2 \\ 2 \\ 1 \end{bmatrix}$

$\lambda_2 = i \quad (A - iI) \vec{u} = \vec{0}$

$\begin{bmatrix} -5-i & 5 & 4 \\ -8 & 7-i & 6 \\ 1 & 0 & -i \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & -i \\ 0 & 1 & 1-i \\ 0 & 0 & 0 \end{bmatrix}$

$u_1, u_2: \text{leading}$ $\vec{u} = u_3 \begin{bmatrix} i \\ i-1 \\ 1 \end{bmatrix}$
 $u_3 = \text{free}$

$\vec{u}_{\lambda=i} = \text{span} \left\{ \begin{bmatrix} i \\ i-1 \\ 1 \end{bmatrix} \right\}$

$\vec{x}_2(t) = e^{it} \begin{bmatrix} i \\ i-1 \\ 1 \end{bmatrix} = (\cos t + i \sin t) \begin{bmatrix} i \\ i-1 \\ 1 \end{bmatrix} = \begin{bmatrix} -\sin t \\ -\sin t - \cos t \\ \cos t \end{bmatrix} + i \begin{bmatrix} \cos t \\ -\sin t \\ \sin t \end{bmatrix}$
 $\text{Im}(\vec{x}_2)$

So $\vec{x}_{\text{general}} = c_1 e^{2t} \begin{bmatrix} 2 \\ 2 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} -\sin t \\ -\sin t - \cos t \\ \cos t \end{bmatrix} + c_3 \begin{bmatrix} \cos t \\ -\sin t \\ \sin t \end{bmatrix}$ $\text{Re}(\vec{x}_2(t))$
 Two real solutions